Progress Check

1. Every section of a sphere by a plane is a _____.
2. The centre of a great circle is at the _____ of the sphere.
3. The difference between the T.S.A. and C.S.A. of hemisphere is _____.
4. The ratio of surface area of a sphere and C.S.A. of hemisphere is _____.
5. A section of the sphere by a plane through any of its great circle is _____.

Example 7.10 If the base area of a hemispherical solid is 1386 sq. metres, then find its total surface area?

Solution Let r be the radius of the hemisphere.

$$\text{Given that, base area } = \pi r^2 = 1386 \text{ sq. m}$$

$$\text{T.S.A. } = 3\pi r^2 \text{ sq.m}$$

$$= 3 \times 1386 \quad = 4158$$

Therefore, T.S.A. of the hemispherical solid is 4158 m^2.

Note

For finding the C.S.A. and T.S.A. of a hollow sphere, the formulla for finding the surface area of a sphere can be used.

Thinking Corner

1. Shall we get a hemisphere when a sphere is cut along the small circle?
2. T.S.A of a hemisphere is equal to how many times the area of its base?
3. How many hemispheres can be obtained from a given sphere?

Example 7.11 The internal and external radii of a hollow hemispherical shell are 3 m and 5 m respectively. Find the T.S.A. and C.S.A. of the shell.

Solution Let the internal and external radii of the hemispherical shell be r and R respectively.

Given that, $R = 5$ m, $r = 3$ m

$$\text{C.S.A. of the shell } = 2\pi(R^2 + r^2) \text{ sq. units}$$

$$= 2 \times \frac{22}{7} \times \left(25 + 9\right) = 213.71$$

$$\text{T.S.A. of the shell } = \pi(3R^2 + r^2) \text{ sq. units}$$

$$= \frac{22}{7}\left(75 + 9\right) \quad = 264$$

Therefore, C.S.A. $= 213.71$ m^2 and T.S.A. $= 264$ m^2.

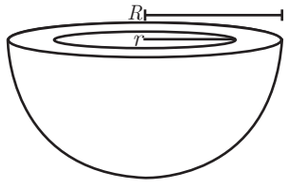


Fig. 7.19

Example 7.12 A sphere, a cylinder and a cone are of the same height which is equal to its radius, where as cone and cylinder are of same height. Find the ratio of their curved surface areas.

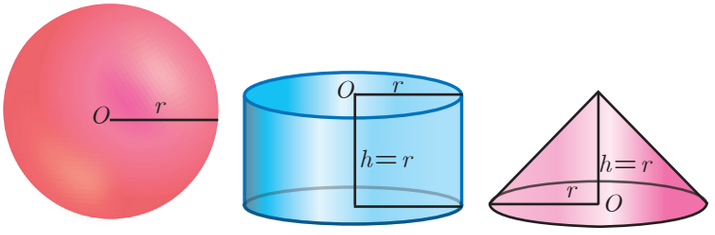


Fig. 7.20

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